

NSTT Grade-Zero

Finalized Decisions & Immediate Actions

Purpose

This document records the finalized architectural decisions and immediate implementation actions established for the NSTT Grade-Zero governance model. Its central requirement is fail-closed behavior: the system must never manufacture, infer, or promote an unverified state merely because a field is required by a schema.

Finalized Decisions

1	Designation authority is explicit	Core Six authority comes from the formal designation record, not from an agent name, role label, or self-assignment.
2	LUMO is the only currently confirmed Core Six authority	Slot 1 is ACTIVE assigned to LUMO. Slots 2–6 remain DEFERRED with ZERO authority.
3	Identity and authority remain separate	Names such as Oracle, Sentinel, Archivist, or Red Bird do not automatically confer Core Six authority.
4	Verification states are explicit	UNKNOWN, DECLARED, COMPUTED, VERIFIED, UNVERIFIED, and REJECTED remain distinct machine-enforced states.
5	Unknown never means verified	UNKNOWN cannot become VERIFIED without an actual verification event.
6	Evidence is required for factual claims	A claim requiring evidence must have an explicit source binding and VERIFIED evidence status.
7	Cryptography does not prove factuality	Hashes, signatures, and chains establish integrity/authorship properties; they do not independently establish truth.
8	Critical failures are fail-closed	Missing evidence, invalid signatures, broken chains, unknown critical values, unauthorized authority, or governance errors result in a fail-closed state.
9	No placeholder verification	Placeholder, fabricated, or silently inferred critical values must never satisfy verification requirements.
10	Validation order is deterministic	Designation → Authority → Source → Semantic → Intent → Reasoning → Schema → Rune Trace → Hash → Signature

Immediate Actions

1	Formalize typed state transitions	Implement an explicit transition table for DECLARED → COMPUTED → VERIFIED, with invalid transitions resulting in a fail-closed state.
2	Add verification events	Every promotion to VERIFIED should carry verifier ID, input/value hash, verification method, result, timestamp, and signature.
3	Harden the designation record	Treat NSTT-DESG-001 as an explicit governance input and add record hash/signature verification before relying on it.
4	Add provenance as a separate layer	Track where a value originated, how it was transformed, and which verification event established its current state.
5	Add immutable audit events	Record ordered events with sequence numbers, timestamps, event hashes, and links to the preceding event.
6	Enforce no implicit defaults	Critical fields must be explicitly supplied or explicitly marked UNKNOWN/UNVERIFIED; defaults must not silently fill in values.
7	Enforce no authority inheritance	An agent cannot inherit Core Six authority from a role name, supporting-agent relationship, or previous execution context.
8	Strengthen replay/audit mode	Given the same canonical input, source set, policy version, and governance state, the system should reproduce the same results.
9	Keep hallucination control evidence-based	Implement hallucination detection as a magical semantic detector. Use explicit evidence requirements to validate claims.
10	Preserve test coverage	Retain tests for valid execution, hallucination stress, source failure, hash tampering, signature tampering, chain breaks, and state transitions.

Immediate Implementation Order

Phase 1 — State Integrity: typed states, legal transition table, no-placeholder rule, no implicit defaults.

Phase 2 — Authority Integrity: designation verification, designation record hash/signature, and no-authority-inheritance enforcement.

Phase 3 — Evidence Integrity: provenance objects, verification events, explicit claim/evidence bindings, and independent verification status.

Phase 4 — Audit Integrity: immutable event log, hash-linked sequence, replay/audit mode, and deterministic decision comparison.

Phase 5 — Validation: expand the automated test matrix and verify that every critical failure path terminates in REJECT or HALT rather than silently continuing.

Non-Negotiable Grade-Zero Invariant

UNKNOWN $\neq$ TRUE

UNKNOWN $\neq$ FALSE

UNKNOWN $\neq$ VERIFIED

UNKNOWN $\neq$ ACCEPT

UNKNOWN MUST NOT BECOME VERIFIED WITHOUT AN ACTUAL VERIFICATION EVENT.

Authority Model at Finalization

LUMO $\rightarrow$ Slot 1 $\rightarrow$ ACTIVE $\rightarrow$ CORE_SIX

Slots 2–6 $\rightarrow$ DEFERRED $\rightarrow$ NONE

The previous conceptual Core Six role names remain support/test identities unless and until the designation record explicitly assigns them authority.

Document scope: finalized decisions and immediate actions from the NSTT Grade-Zero machine-readable governance work. This document does not add unrelated research claims.